

Java String Case Study – Student Version (Questions with Sample Input & Expected Output)

Sample Employee Record

Field	Value
Employee Name	John Smith
Email	John.Smith@ABC.com
Employee ID	EMP-1023
Password	John@123

1. Remove leading/trailing spaces

Sample Input: Employee Name: ' John Smith '

Expected Output: John Smith

2. Convert email to lowercase

Sample Input: John.Smith@ABC.com

Expected Output: john.smith@abc.com

3. Validate email

Sample Input: John.Smith@ABC.com

Expected Output: Valid

4. Validate Employee ID

Sample Input: EMP-1023

Expected Output: Valid

5. Generate username

Sample Input: John Smith

Expected Output: john.smith

6. Count words

Sample Input: John Smith

Expected Output: 2

7. Check password strength

Sample Input: John@123

Expected Output: Strong

8. Mask email

Sample Input: john.smith@abc.com

Expected Output: jo*****@abc.com

9. Find duplicate emails

Sample Input: john@gmail.com, mary@gmail.com, john@gmail.com

Expected Output: Duplicate: john@gmail.com

10. Extract domain

Sample Input: john.smith@abc.com

Expected Output: abc.com

11. Count vowels

Sample Input: John Smith

Expected Output: 2

12. Count uppercase/lowercase/digits/special

Sample Input: John@123

Expected Output: Uppercase=1, Lowercase=3, Digits=3, Special=1

13. Reverse employee name

Sample Input: John Smith

Expected Output: htimS nhoJ

14. Palindrome check

Sample Input: Madam

Expected Output: Palindrome

15. First non-repeated character

Sample Input: swiss

Expected Output: w

16. Most frequent character

Sample Input: programming

Expected Output: g

17. Replace spaces with underscores

Sample Input: John Smith

Expected Output: John_Smith

18. Compare two names

Sample Input: Java & new String("Java")

Expected Output: equals() = true, == = false

19. Sort names

Sample Input: John, Alice, David

Expected Output: Alice, David, John

20. Generate employee code

Sample Input: John Smith, EMP-1023

Expected Output: JS1023